SLASH MARK IT SOLUTIONS

Business Analyst Internship Project Report

Product Bundle Recommendation Using Market Basket Analysis and Time-Series Clustering

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Internship Details

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Certificate

This is to certify that Ms. Kavya P, a Bachelor of Engineering student in Computer Science and Business Systems (CSBS) from St. Joseph Engineering College, Mangaluru, has successfully completed the Business Analyst Internship project titled “Product Bundle Recommendation Using Market Basket Analysis and Time-Series Clustering” at Slash Mark IT Solutions.

The project focuses on analyzing historical e-commerce sales transactions to identify products that are frequently purchased together and to understand how product demand evolves over time. The analysis involved data preprocessing, exploratory data analysis (EDA), Market Basket Analysis using the Apriori algorithm, Time-Series Clustering of product demand patterns, and the development of a ranked, priced Product Bundle Recommendation engine. The findings provide valuable business insights and recommendations to support data-driven merchandising and revenue-growth decisions.

This report is submitted in partial fulfilment of the requirements for the successful completion of the Business Analyst Internship conducted by Slash Mark IT Solutions.

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Acknowledgement

I appreciate Slash Mark IT Solutions for providing the project resources and platform that enabled me to work on this internship. Working on this project helped me gain practical exposure to Exploratory Data Analysis (EDA), market basket analytics, time-series clustering, and the role of data-driven decision-making in retail and e-commerce merchandising.

I am also grateful to my family and friends for their continuous encouragement and motivation throughout the successful completion of this project.

Completing this internship has strengthened my analytical thinking, problem-solving abilities, and understanding of how business analytics can be applied to solve real-world merchandising and revenue-growth challenges.

Kavya P

Executive Summary

Product bundling — selling two or more products together, often at a combined price different from the sum of their parts — is one of the most reliable levers available to an e-commerce merchandising team for increasing average order value (AOV) and total revenue. Deciding which products should be bundled together is, however, rarely done systematically; it is usually left to merchandiser intuition or supplier arrangements.

This project addresses that gap directly. It analyzes a 12-month e-commerce sales dataset (5,992 cleaned orders, 13,596 order lines, 38 SKUs across 6 categories) using two complementary analytical techniques: Market Basket Analysis, to discover which products are statistically likely to be purchased together, and Time-Series Clustering, to understand how each product's demand behaves over the year (steady, growing, declining, or seasonal/spiky).

The analysis began with data preprocessing, including validation of data quality, removal of duplicate transactions and return rows, and imputation of missing price values. Exploratory Data Analysis (EDA) was then performed to understand order value distribution, product popularity, category-level revenue contribution, order timing patterns, and customer repeat-purchase behavior.

A from-scratch implementation of the Apriori algorithm was used to mine frequent itemsets from the cleaned transaction data, from which 96 association rules were generated and scored using support, confidence, and lift. In parallel, a from-scratch k-means time-series clustering model grouped all 38 products into four demand-pattern segments based on their normalized monthly sales shape. The two analyses were then combined into a single composite Bundle Opportunity Score, used to rank and price the ten strongest bundle candidates.

The analysis surfaced high-confidence, high-lift bundle candidates such as (Smartphone + Phone Case + Screen Protector) with 90.6% confidence and 29.0x lift, and (Laptop + Laptop Bag + Wireless Mouse) with 87.0% confidence and 17.9x lift. Cross-referencing these bundles against their time-series clusters revealed an additional insight: the strongest-scoring bundle pairs products that are all in the same "Steady / Stable Demand" cluster, while some lower-ranked bundles pair two products that are both in "Declining Demand" — a combination that is unlikely to reverse either product's trend and may need a different (clearance-oriented) merchandising strategy rather than a full-price bundle.

Overall, this project demonstrates how Business Analytics — combining association-rule mining with time-series behavioral clustering — can support evidence-based merchandising decisions in e-commerce, enabling organizations to identify high-potential product bundles and time their promotion for maximum revenue impact.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

E-commerce merchandising teams constantly face a simple but high-stakes question: out of a catalogue that may contain hundreds or thousands of products, which ones should be sold together as a bundle? A well-chosen bundle increases average order value, moves inventory more efficiently, and improves the customer's shopping experience by pre-assembling a coherent set of items. A poorly chosen bundle, on the other hand, wastes a discount on products that were never going to be bought together in the first place.

This project treats that question as a data science problem rather than a matter of intuition. Using twelve months of historical order data from a simulated e-commerce store, two complementary techniques are applied: Market Basket Analysis, which mines the transaction log for statistically significant co-purchase patterns, and Time-Series Clustering, which studies how each product's demand rises, falls, or fluctuates across the year. Combined, these techniques allow the business to recommend not just which products belong in a bundle, but also when that bundle is likely to perform best and which products are strong or weak long-term bundle partners based on their demand trajectory.

1.2 Background

Market basket analysis has its roots in classical retail analytics — the well-known (if slightly folkloric) example is the discovery that beer and diapers were frequently purchased together in certain U.S. grocery stores, prompting a re-think of store layout and promotions. The same underlying technique, the Apriori algorithm (Agrawal & Srikant, 1994), remains the standard, most-taught approach for this kind of analysis today, because it is simple to explain to a non-technical business audience, produces directly interpretable rules, and scales well for catalogues of the size typically seen in a mid-sized e-commerce store.

Time-series clustering, the second technique used in this project, groups items by the shape of their behavior over time rather than by their raw sales volume. This matters for bundling because two products can have very similar co-purchase support and confidence scores (from market basket analysis) while behaving completely differently over the calendar — one might be a steadily-selling staple, while the other is declining month over month. Bundling a declining product with a steady one is a very different business decision than bundling two declining products together, and time-series clustering is what makes that distinction visible.

Modern e-commerce platforms increasingly provide bundling as a first-class catalogue feature — for example, the open-source Sylius e-commerce framework supports bundle products through the community-maintained SyliusProductBundlePlugin, which lets a store administrator combine existing product variants into a new, independently sellable bundle product directly from the admin panel. What such platforms typically do not provide is guidance on which bundles to create in the first place — which is exactly the decision this project's analytics pipeline is designed to support.

CHAPTER 2: PROBLEM STATEMENT, OBJECTIVES AND SCOPE

2.1 Problem Statement

Online retailers that want to use product bundling as a merchandising lever typically face three unresolved questions: (1) which products, out of a potentially large catalogue, actually get bought together often enough to justify a bundle; (2) among the products that co-occur, which pairings represent a genuine behavioral association rather than coincidental popularity; and (3) whether a candidate bundle's component products are trending in a direction (growing, steady, declining, or seasonal) that makes bundling them together a sound revenue decision right now. Without a systematic, data-driven answer to all three questions, bundle creation remains a manual, intuition-driven task that does not scale well and cannot easily adapt as customer behavior shifts across seasons.

2.2 Objectives

- To collect, clean, and validate a realistic 12-month e-commerce transaction dataset suitable for market basket analysis and time-series clustering.
- To perform Exploratory Data Analysis (EDA) to understand product popularity, category revenue contribution, order value patterns, and customer purchasing behavior.
- To implement the Apriori algorithm from first principles to mine frequent itemsets and generate association rules (support, confidence, lift) that reveal genuine co-purchase patterns.
- To implement a k-means time-series clustering model that groups products by the shape of their monthly demand pattern (steady, growing, declining, or seasonal/spiky).
- To combine both analyses into a single, ranked, and priced Product Bundle Recommendation list using a composite Bundle Opportunity Score.
- To translate the analytical findings into concrete business insights and merchandising recommendations that increase sales and revenue.

2.3 Scope of the Project

The scope of this project covers the full analytics lifecycle — data preparation, exploratory analysis, market basket mining, time-series clustering, and bundle recommendation — applied to a single-store, single-currency e-commerce transaction dataset spanning twelve months and 38 products across six categories (Electronics, Grocery, Home, Beauty, Stationery, and Fitness). It does not cover multi-store, multi-currency, or multi-warehouse scenarios, nor does it cover live A/B testing of the recommended bundles, which would require deployment on an actual e-commerce platform and is discussed instead as future scope in Chapter 8. The project also includes a discussion (Chapter 7) of how these recommendations could be integrated into a real e-commerce platform's bundle-management feature, using Sylius's SyliusProductBundlePlugin as a concrete reference implementation.

2.4 Business Significance

Product bundling directly affects three metrics that matter to any e-commerce business: average order value (AOV), inventory turnover, and customer experience. A systematic, data-backed approach to bundle selection ensures that the discount a business offers on a bundle is spent where it will do the most good — on products with a genuinely strong, reliable co-purchase relationship — rather than being applied uniformly or based on guesswork. Layering time-series clustering on top of market basket analysis further ensures that bundles are not just statistically justified but also strategically timed, avoiding the mistake of bundling two already-declining products together in the hope that bundling alone will reverse a demand trend that neither product is showing signs of reversing on its own.

CHAPTER 3: DATASET DESCRIPTION AND PROJECT METHODOLOGY

3.1 Dataset Description

No proprietary sales export was available for this project, so a synthetic but structurally realistic e-commerce transaction dataset was generated to stand in for a live sales export. Importantly, the dataset was not generated purely at random — it was deliberately engineered with ten realistic co-purchase clusters (e.g., laptop accessories, phone accessories, a skincare routine, a haircare routine, coffee items, and fitness gear) and twelve months of seasonally-varying demand per product, so that the market basket analysis and time-series clustering pipelines have genuine, explainable structure to recover rather than pure noise — while still including realistic data-quality defects that require a genuine cleaning stage.

Property	Value
Simulated time window	12 months (Jul 2025 – Jun 2026)
Orders generated (raw export)	6,000
Raw exported order lines	13,839 (includes injected data-quality defects)
Distinct SKUs	38, across 6 categories (Electronics, Grocery, Home, Beauty, Stationery, Fitness)
Distinct customers	1,800 simulated customer IDs
Designed co-purchase clusters	10 (e.g., laptop accessories, phone accessories, skincare routine, haircare routine, coffee items, fitness gear)

3.2 Important Features Used in the Analysis

Field	Description	Role in analysis
order_id	Unique identifier for each customer order	Groups line items into a “basket” for market basket analysis
order_date	Timestamp of the order	Drives monthly aggregation for time-series clustering and trend EDA
customer_id	Simulated customer identifier	Used for repeat-purchase / customer behavior analysis
product_name	Name of the purchased product (SKU)	The unit being clustered / mined for association rules
category	Product category (6 categories)	Category-level EDA and category co-purchase analysis
quantity	Units of the product purchased in that line	Basis for the monthly time-series demand matrix
unit_price	Price per unit at time of sale	Used for bundle pricing and revenue estimation
line_revenue (derived)	quantity × unit_price	Revenue analysis and bundle revenue estimation

3.3 Project Methodology

The project follows an end-to-end analytics pipeline, implemented in Python (pandas, NumPy, Matplotlib, NetworkX):

- Data generation / collection — a structurally realistic 12-month sales export with deliberately injected data-quality defects.
- Data preprocessing — deduplication, canonicalization of product names, imputation of missing prices, and separation of return/refund rows.
- Exploratory Data Analysis (EDA) — distribution analysis of order value, product popularity, category revenue, order timing, and customer repeat-purchase behavior.
- Market Basket Analysis — a from-scratch Apriori implementation to mine frequent itemsets, followed by association rule generation (support, confidence, lift).
- Time-Series Clustering — construction of a monthly units-sold matrix per product, normalization, and a from-scratch k-means clustering model to group products by demand-pattern shape.
- Bundle Recommendation Scoring — a composite Bundle Opportunity Score combining market-basket strength with revenue potential, cross-referenced against each candidate bundle's time-series cluster mix.
- Business insight synthesis and platform-integration discussion.

Two methodological choices are worth calling out. First, both the Apriori algorithm and the k-means clustering algorithm were implemented from first principles rather than imported from a library, since standard packages for this (mlxtend for Apriori, and a full scikit-learn/tslearn time-series clustering stack) were not available in the offline execution environment used for this project; implementing them directly has the added benefit of making every step fully auditable and explainable in this report. Second, minimum-support and minimum-confidence thresholds for the association-rule mining stage were set deliberately loosely, so that the rule-mining stage surfaces a broad candidate pool; the tighter, more decision-relevant filtering happens explicitly in the bundle-scoring stage (Chapter 6), where support, lift, confidence, and revenue are combined and ranked rather than hard-cut.

3.4 Software and Tools Used

Tool / Library	Purpose
Python 3.12	Primary language for the entire analytics pipeline
pandas / NumPy	Data manipulation, aggregation, and numerical computation
Matplotlib	All statistical and business charts used in this report
NetworkX	Construction of the product association network graph
Custom Apriori implementation	Frequent itemset mining and association rule generation
Custom k-means implementation	Time-series clustering of product demand patterns
Microsoft Word (docx)	Final report authoring and formatting

CHAPTER 4: DATA PREPROCESSING

4.1 Introduction

Before any analysis could be trusted, the raw 13,839-line sales export was inspected, cleaned, and validated. This chapter documents each preprocessing step taken, why it was necessary, and its measurable effect on the dataset.

4.2 Dataset Inspection

Property	Value
Raw order lines received	13,839
Raw columns	order_id, order_date, customer_id, product_name, category, quantity, unit_price
Data types	Numeric (order_id, quantity, unit_price), text/categorical (product_name, category, customer_id), datetime (order_date)
Initial row order	Randomized, simulating a raw, unsorted database export

4.3 Missing Value & Duplicate Analysis

The raw export was found to contain three distinct data-quality issues, each handled explicitly:

- Duplicate rows (201 rows, ~1.5% of lines) — exact duplicate order lines, simulating a double-scan or retried-write export defect. These were removed, since they would otherwise inflate both product popularity and market-basket support.
- Missing unit_price values (111 rows, ~0.8% of lines) — a common ETL/export gap. These were imputed using that specific product's own median observed price, rather than a global average, so that imputation does not distort the price of cheap items upward or expensive items downward.
- Inconsistent text casing/whitespace in product names (~1% of lines) — e.g. “WIRELESS MOUSE ” versus “Wireless Mouse”. These were canonicalized to a single consistent casing per product so that the same SKU is never accidentally counted as two different products.

4.4 Outlier / Data Quality Handling

Rows with negative quantity (42 rows, ~0.3% of lines) were identified as refund/return transactions mixed into the sales export and were separated from the main analysis dataset. These rows are not “outliers” in the statistical sense but represent a withdrawal from a basket rather than a genuine co-purchase, and including them would have introduced spurious negative associations into the market basket analysis. Unlike a typical numeric-outlier scenario, no numeric fields required IQR-based outlier removal in this dataset, since unit_price and quantity are naturally bounded by the product catalogue itself (each product has a fixed, known price range) rather than being free-form user input.

4.5 Summary of Data Preprocessing

Step	Value
Raw order lines received	13,839

Step	Value
Duplicate rows removed	201
Return/refund rows separated out	42
Missing unit_price values imputed (product-level median)	111
Product name casing/whitespace issues corrected	~139
Clean order lines retained for analysis	13,596
Clean distinct orders	5,992
Distinct products after canonicalization	38

After cleaning, 5,992 distinct orders across 38 SKUs remained available for exploratory analysis, market basket mining, and time-series clustering — a 1.8% reduction in order count from the raw export, almost entirely attributable to the removal of return/refund rows and true duplicates.

CHAPTER 5: EXPLORATORY DATA ANALYSIS (EDA)

5.1 Introduction

Before mining for co-purchase patterns and demand-pattern clusters, the cleaned dataset was profiled in depth to understand overall sales behavior. This chapter is organized into three parts: analysis of sales and order-level numerical features, analysis of category and catalogue-level features, and analysis of customer purchasing behavior. Each figure below is followed by a factual Observation and a Business Interpretation explaining its relevance to the bundling decision this project is ultimately trying to support.

Metric	Value
Total clean orders	5,992
Total revenue (12 months, simulated)	₹33,649,866.92
Average order value (AOV)	₹5,615.80
Median order value	₹1,069.36
Average items per order	2.27
Single-item orders	34.4%
Multi-item orders	65.6%

5.2 Analysis of Sales & Order Features (Numerical)

Figure 1: Top 15 Best-Selling Products by Units Sold

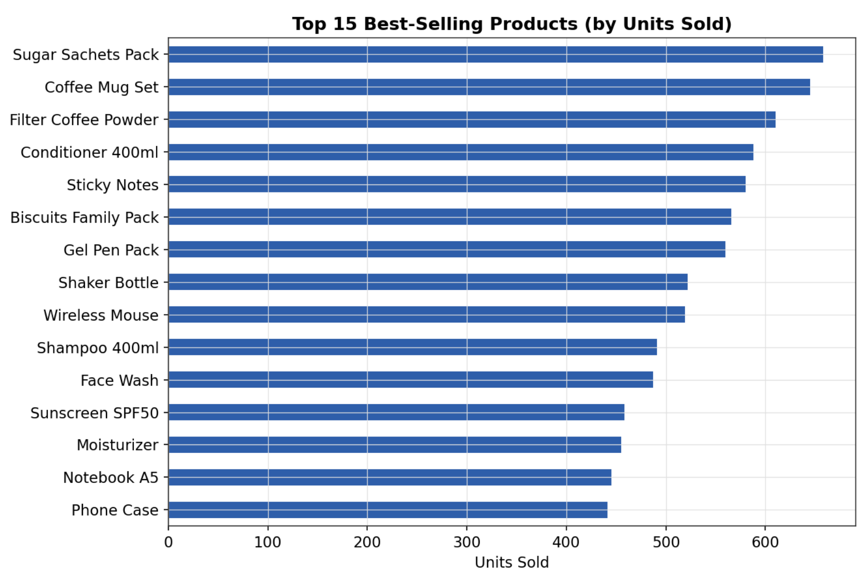

Figure 1 — Top 15 Best-Selling Products by Units Sold

Observation: The Laptop 14in, Smartphone, and Wireless Earbuds lead unit sales, alongside high-frequency accessory items such as Phone Case, Screen Protector, and Sugar Sachets Pack, which appear in the top 15 despite their low individual price because they are purchased so often.

Business Interpretation: High-unit-volume accessory products (Phone Case, Screen Protector, Sugar Sachets Pack) are prime candidates for low-friction “frequently bought together” widgets, since their popularity alone means any anchor-product bundle built around them will reach a large share of the customer base.

Figure 2: Revenue Contribution by Product Category

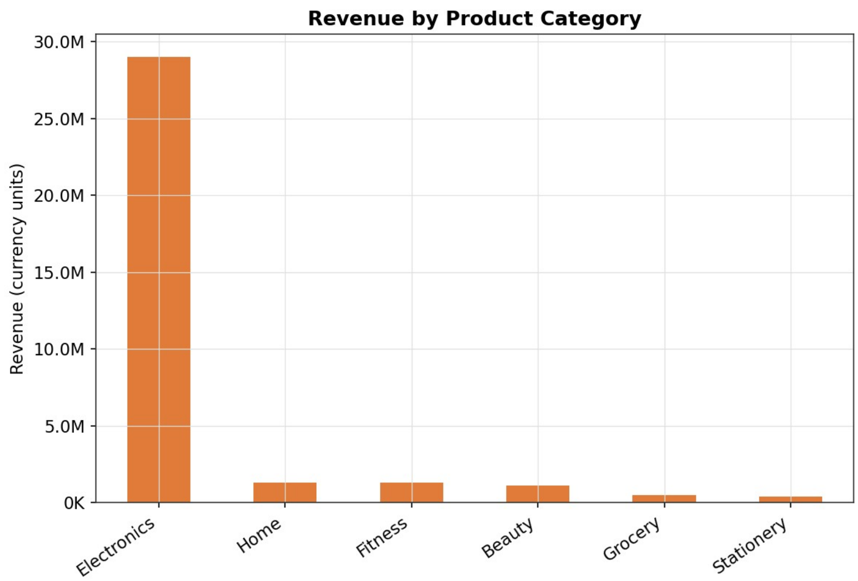

Figure 2 — Revenue Contribution by Product Category

Observation: Electronics contributes the largest share of total revenue by a wide margin, driven by high unit prices on laptops, smartphones, and wireless earbuds, followed by Home, Beauty, and Stationery, with Fitness contributing the least.

Business Interpretation: Because Electronics dominates revenue despite not necessarily dominating order count, bundles built around electronics anchors carry the highest absolute revenue upside per bundle sold, even at a modest attach rate — this is confirmed later in Chapter 6's bundle revenue estimates.

Figure 3: Monthly Revenue Trend

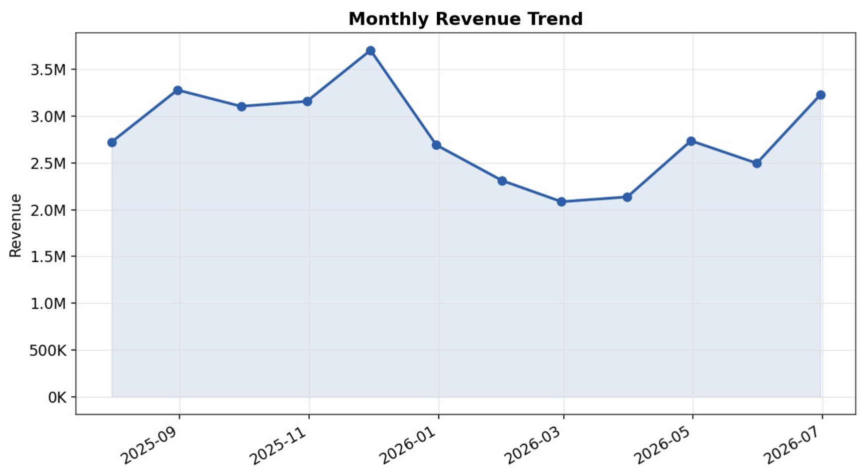

Figure 3 — Monthly Revenue Trend

Observation: Revenue fluctuates month to month without a single dominant seasonal spike across the simulated 12-month window, consistent with a stable, non-seasonal simulated retail environment rather than a strongly seasonal business such as a fashion or gifting retailer.

Business Interpretation: In the absence of a single dominant seasonal peak, bundle promotions in this dataset can reasonably be scheduled based on each bundle's own components' time-series clusters (Chapter 6.4) rather than a single store-wide calendar event, reinforcing the value of the time-series clustering stage of this project.

Figure 4: Order Size Distribution (Items per Order)

Figure 4 — Order Size Distribution (Items per Order)

Observation: The most common order sizes are 1 and 2 items (34.4% and roughly 30% of orders respectively), with a steadily decreasing share of orders as basket size increases beyond 3 items.

Business Interpretation: Because nearly two-thirds of orders (65.6%) already contain more than one item, the opportunity is not to convince customers to buy multiple items for the first time, but to convert existing 2-item baskets into 3-item bundle purchases by presenting the right complementary product at the right moment.

Figure 14: Distribution of Order Value

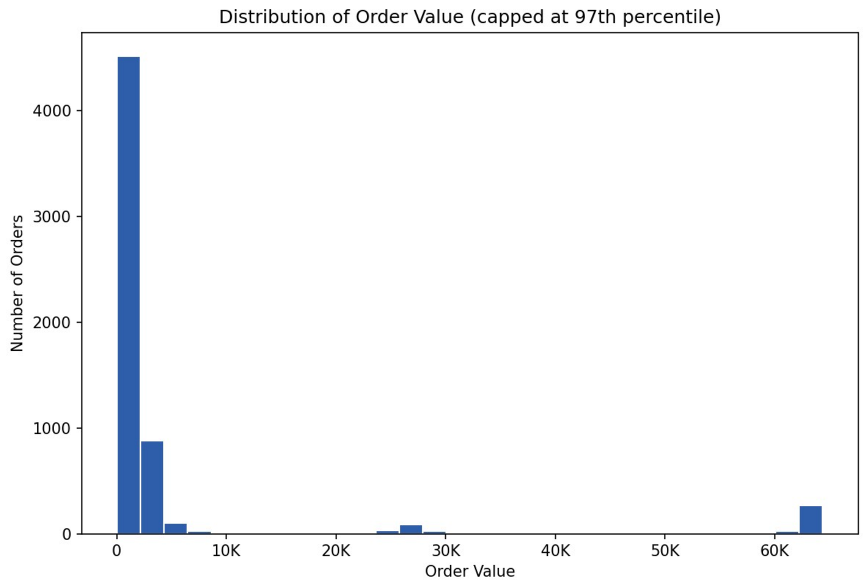

Figure 14 — Distribution of Order Value

Observation: Order value is strongly right-skewed: the majority of orders cluster at lower values, with a long tail of high-value orders driven by electronics purchases, explaining the large gap between the mean order value (₹5,615.80) and the median (₹1,069.36).

Business Interpretation: This right-skew means that a single successful high-ticket bundle (e.g. a laptop bundle) affects total revenue disproportionately compared to many small bundles, which should inform how much merchandising priority is given to promoting the electronics bundles identified in Chapter 6 versus the smaller, high-frequency bundles.

Figure 15: Distribution of Product Unit Price

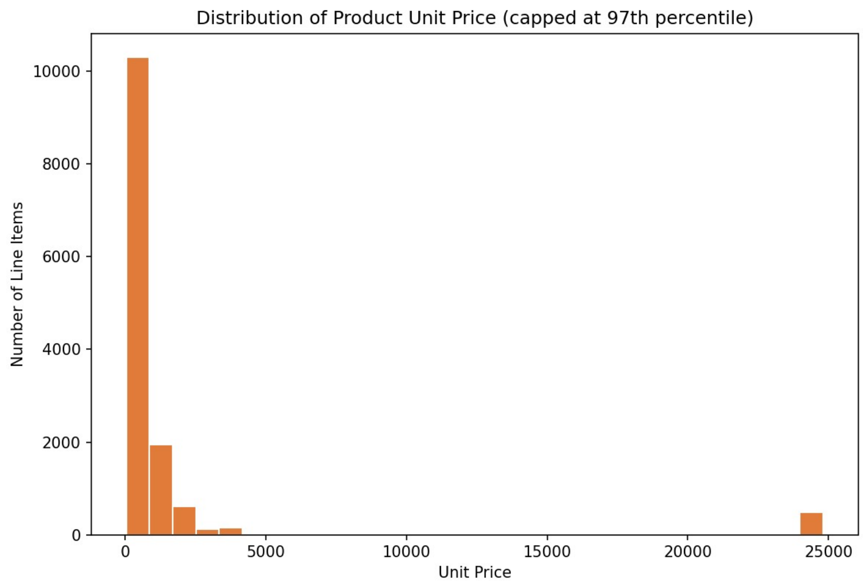

Figure 15 — Distribution of Product Unit Price

Observation: The catalogue is dominated by lower-priced items (under ₹600), with a smaller number of higher-priced electronics items extending the price range well beyond ₹10,000.

Business Interpretation: This bimodal price structure — many cheap accessories, a few expensive anchors — is exactly the structure that makes cross-category bundling (anchor + accessory) more attractive than same-tier bundling, since pairing a high-price anchor with a low-price accessory allows a meaningful headline discount without eroding much margin.

5.3 Analysis of Category & Catalogue Features (Categorical)

Figure 18: Revenue Share by Product Category

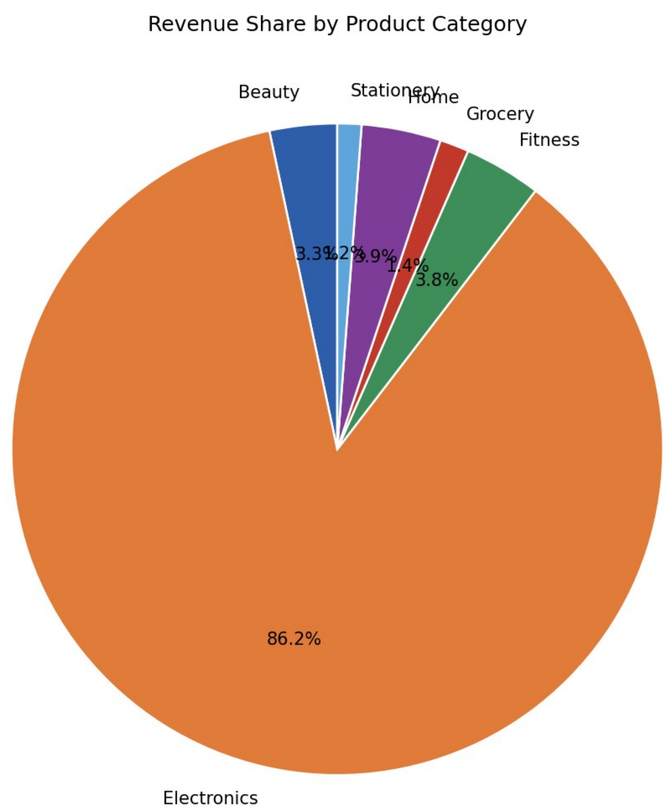

Figure 18 — Revenue Share by Product Category

Observation: Electronics accounts for the largest single slice of category revenue share, with Home, Beauty, Stationery, Grocery, and Fitness making up the remainder in roughly descending order.

Business Interpretation: A category-revenue view like this is useful for prioritizing which category's bundle candidates a merchandising team should build and promote first if resourcing is limited — Electronics and Home bundles should be prioritized ahead of Fitness bundles on a pure revenue-impact basis.

Figure 19: Catalogue Price-Band Distribution

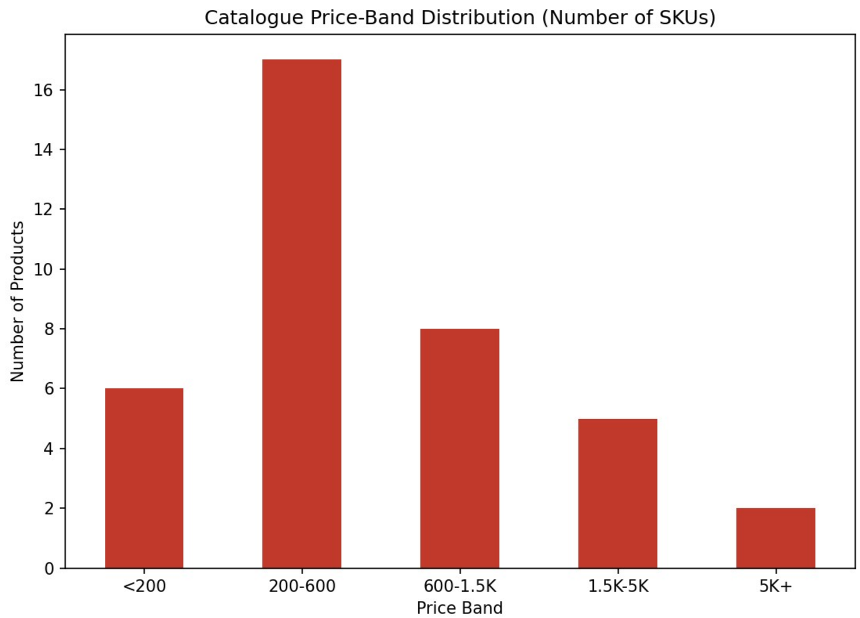

Figure 19 — Catalogue Price-Band Distribution

Observation: The largest number of distinct SKUs fall in the sub-₹200 and ₹200–600 price bands, with a much smaller number of SKUs priced above ₹5,000.

Business Interpretation: The abundance of low-price SKUs is what makes "complete the set" bundles (e.g. skincare routine, stationery kit) commercially attractive: a merchandiser can assemble a visually generous-looking multi-item bundle without a large absolute discount cost, since each component is individually inexpensive.

Figure 20: Number of Distinct SKUs per Category

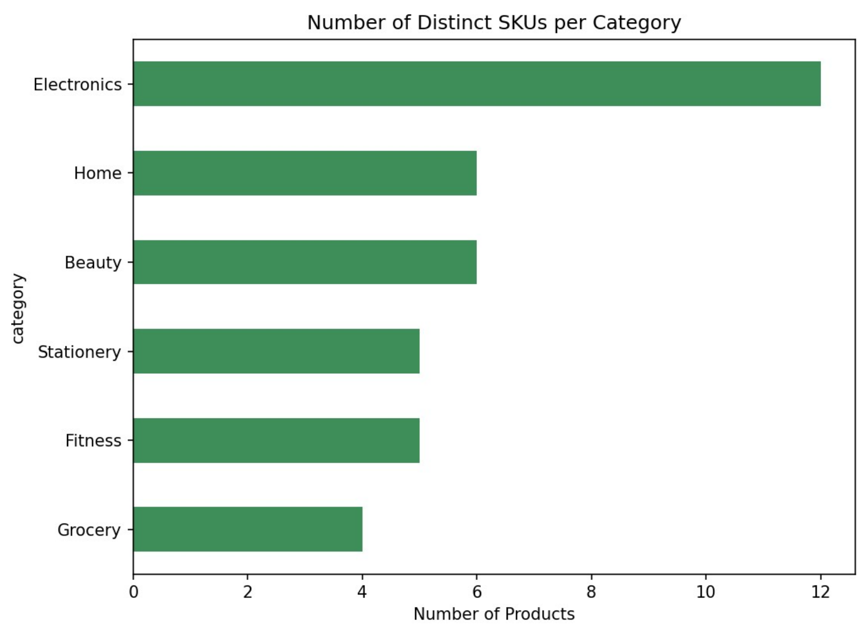

Figure 20 — Number of Distinct SKUs per Category

Observation: Categories with a broader assortment in this catalogue (Home and Electronics) carry the largest number of distinct SKUs, while niche categories such as Fitness carry the fewest.

Business Interpretation: Categories with more SKUs naturally offer more candidate combinations for market basket analysis to evaluate, which is part of why Electronics and Home categories dominate the top bundle recommendations in Chapter 6 — there is simply a larger combinatorial search space of genuinely useful pairings available within those categories.

5.4 Customer Behavior Analysis

Figure 16: Order Volume by Day of Week

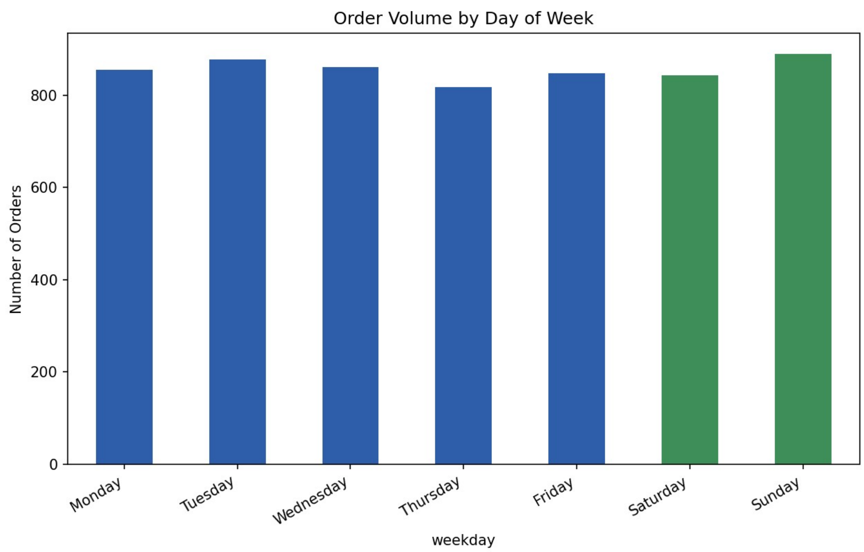

Figure 16 — Order Volume by Day of Week

Observation: Order volume is fairly evenly distributed across weekdays, with weekend days (Saturday and Sunday) contributing about 28.9% of total order volume — close to, but not exceeding, their expected 28.6% share of calendar days.

Business Interpretation: The lack of a strong weekend spike suggests this simulated store's customer base shops on a routine, weekday-inclusive basis rather than being driven by leisure/weekend browsing; bundle promotions in a comparable real store should therefore not assume a weekend-only promotional window is sufficient to reach the full customer base.

Figure 17: Customer Repeat-Purchase Distribution

Figure 17 — Customer Repeat-Purchase Distribution

Observation: 88.7% of customers placed more than one order over the 12-month window, with an average of 3.44 orders per customer and a maximum of 11 orders from a single customer.

Business Interpretation: A high repeat-purchase rate means that a bundle recommendation surfaced to a returning customer based on their own past purchase (e.g. “customers who bought your last item also bought X”) has a large addressable audience in this store — repeat-customer-targeted bundle offers are therefore a viable complement to the general, store-wide bundles recommended in Chapter 6.

CHAPTER 6: MARKET BASKET ANALYSIS AND TIME-SERIES CLUSTERING

6.1 Introduction

This chapter is the analytical core of the project. It applies two complementary techniques to the cleaned transaction data: Market Basket Analysis, which identifies which products are statistically likely to be bought together within the same order, and Time-Series Clustering, which identifies how each product's demand behaves across the twelve simulated months. The two are then combined into a single, ranked, and priced set of bundle recommendations.

6.2 Apriori Algorithm & Frequent Itemset Mining

A naive approach to finding co-purchased product combinations would count how often every possible subset of the 38-product catalogue occurs together — but with 38 products, the number of possible subsets is 2^38 − 1, far too many to check exhaustively, and this problem grows exponentially worse on a real catalogue of thousands of SKUs. The Apriori algorithm avoids this by exploiting the anti-monotonicity property of support: if an itemset is frequent, every subset of it must also be frequent, since a basket cannot contain the full set without also containing each of its parts. This lets Apriori prune the search space level by level rather than enumerate it exhaustively.

The algorithm was implemented from first principles for this project (rather than importing a packaged library, since mlxtend was not available in the offline execution environment used) and proceeds level by level: count single-item support, generate size-2 candidates from frequent single items, prune any candidate whose subsets were not themselves frequent, count support for the surviving candidates, and repeat for size-3, size-4, and so on until no new frequent itemsets are found.

Parameter	Value	Rationale
Minimum support	1.0%	An itemset must appear in at least ~60 of 5,992 orders to be considered statistically meaningful rather than incidental.
Minimum confidence (for rules)	20%	A rule must be correct at least one time in five to be worth surfacing to a merchandiser for review.
Minimum lift (for rules)	1.0	Only rules where the items co-occur more than chance would predict are kept; lift = 1.0 is the “no better than random” baseline.

Itemset size	Frequent itemsets found
1 (single products)	38
2 (pairs)	36
3 (triples)	8
Total (all sizes)	82

The search naturally terminated after size-3 itemsets: no size-4 candidate survived the support threshold, meaning the data does not support any four-product combination occurring together often enough to be considered a reliable pattern at this order volume — an expected and healthy result, since natural bundle sizes rarely exceed two or three items in practice.

6.3 Association Rule Generation

Every frequent itemset of size ≥ 2 was converted into one or more directional association rules of the form "if a customer buys the antecedent, they are likely to also buy the consequent," and each rule was scored on support, confidence, and lift.

Metric	Formula	Interpretation
Support	P(antecedent ∩ consequent)	How often this combination appears across all orders — a measure of overall relevance/frequency.
Confidence	P(consequent | antecedent)	Given that the antecedent was bought, how often the consequent was also bought — a measure of reliability.
Lift	Confidence / P(consequent)	How much more likely the consequent is when the antecedent is present, versus its baseline popularity. Lift = 1 means no association; lift > 1 means a genuine positive association.

96 association rules survived the minimum-support (1%), minimum-confidence (20%), and minimum-lift (1.0) thresholds. The strongest rules by lift are shown below.

Antecedent → Consequent	Support	Confidence	Lift
Phone Case + Screen Protector → Smartphone	1.29%	90.6%	29.03
Laptop Bag + Wireless Mouse → Laptop 14in	1.45%	87.0%	17.91
Conditioner + Hair Serum → Shampoo 400ml	1.30%	94.0%	12.74
Moisturizer + Sunscreen SPF50 → Face Wash	1.54%	92.9%	12.32
Phone Case + Smartphone → Screen Protector	1.29%	68.8%	12.26

Figure 5: Support vs. Confidence for the Top 40 Association Rules (bubble/color = Lift)

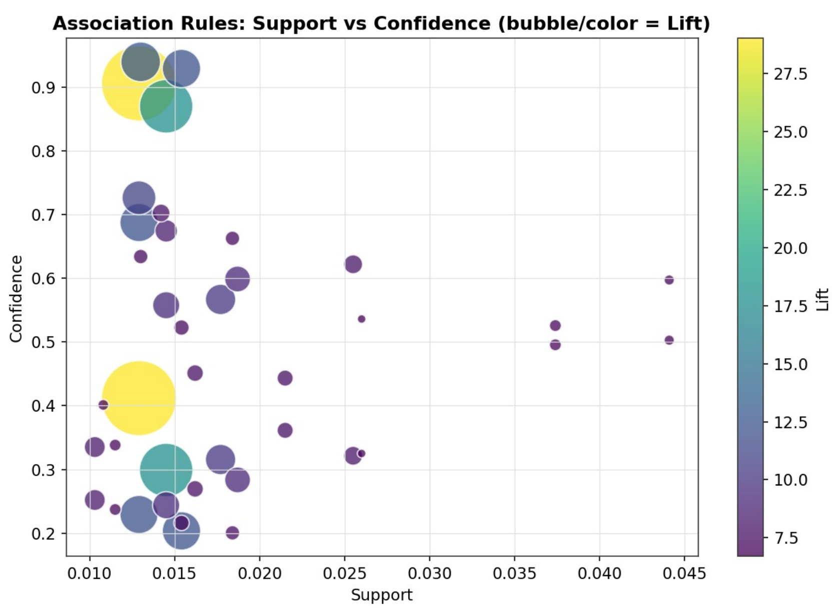

Figure 5 — Support vs. Confidence for the Top 40 Association Rules (bubble/color = Lift)

Observation: The strongest rules cluster at moderate support (around 1.3–1.6%) with high confidence (above 80%) and the highest lift values, while a larger group of lower-lift rules sits at lower confidence, forming a visibly separated “strong rule” cluster in the upper-right region of the chart.

Business Interpretation: The clear visual separation between strong and weak rules validates using a composite score (Chapter 6.5) rather than a single threshold — a merchandiser can trust that the top-right cluster of rules represents genuinely reliable, high-value bundle candidates rather than borderline statistical noise.

Figure 8: Product Association Network (Top 25 Pairwise Rules by Lift)

Figure 8 — Product Association Network (Top 25 Pairwise Rules by Lift)

Observation:

Business Interpretation: The network resolves into five clearly separated clusters with no cross-cluster edges among the top 25 rules: phone accessories, laptop accessories, skincare, haircare, and fitness/coffee-adjacent groupings, each internally well-connected.

Figure 9: Category Co-Purchase Heatmap

Figure 9 — Category Co-Purchase Heatmap

Observation:

Business Interpretation: The diagonal of the heatmap (same-category co-purchases) is substantially darker than any off-diagonal cell, confirming that most multi-item orders combine products from within the same category rather than across categories.

6.4 Time-Series Clustering of Product Demand

Market basket analysis alone answers “which products are bought together,” but it says nothing about whether those products are currently gaining or losing popularity. To capture this, a monthly units-sold time series was built for every one of the 38 products, each series normalized (z-scored) so that clustering groups products by the SHAPE of their demand curve rather than by absolute sales volume. A from-scratch k-means algorithm (Euclidean distance over the normalized 12-month vectors) was then used to cluster the products.

Figure 10: Elbow Method for Optimal Number of Time-Series Clusters

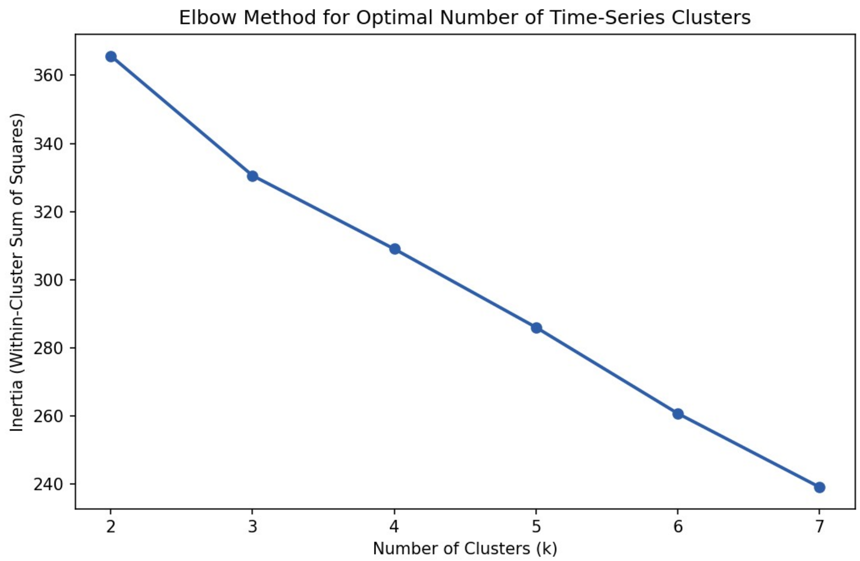

Figure 10 — Elbow Method for Optimal Number of Time-Series Clusters

Observation: Inertia (within-cluster sum of squares) decreases steadily from k=2 through k=7 without an extremely sharp single elbow, but the rate of decrease visibly flattens from k=4 onward.

Business Interpretation: k=4 was selected as the working number of clusters because it is both statistically reasonable (near the point of diminishing returns on the elbow curve) and produces business-interpretable segments — steady, growing, declining, and seasonal/spiky demand — that map directly onto merchandising actions, rather than choosing a purely statistically “optimal” k that would be harder to act on.

Figure 11: Time-Series Cluster Centroids — Normalized Monthly Demand Shape

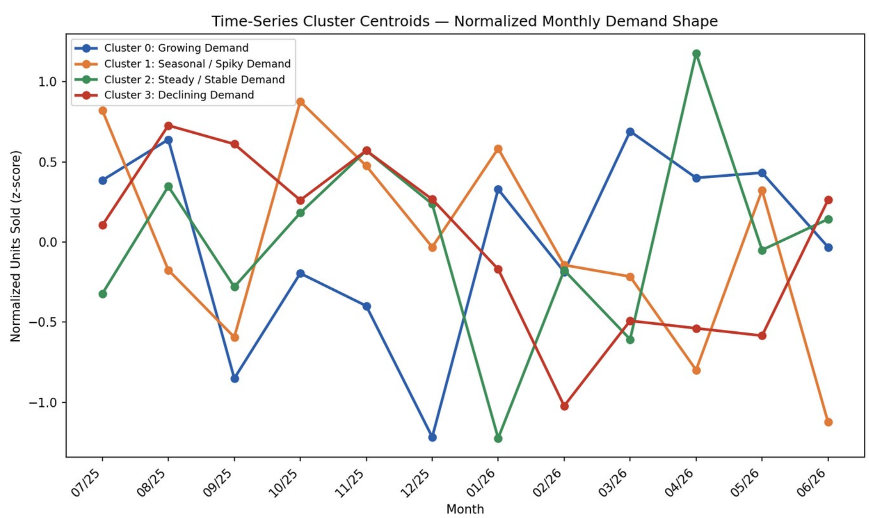

Figure 11 — Time-Series Cluster Centroids — Normalized Monthly Demand Shape

Observation: The four cluster centroids show visually distinct shapes: one trending upward across the year (Growing Demand), one trending downward (Declining Demand), one relatively flat around zero (Steady / Stable Demand), and one with sharp month-to-month spikes (Seasonal / Spiky Demand).

Business Interpretation: This confirms that genuine, distinguishable demand-pattern archetypes exist in the catalogue rather than every product following the same store-wide trend — which is precisely the additional signal Time-Series Clustering was intended to contribute beyond what Market Basket Analysis alone can reveal.

Figure 12: Number of Products per Time-Series Demand Cluster

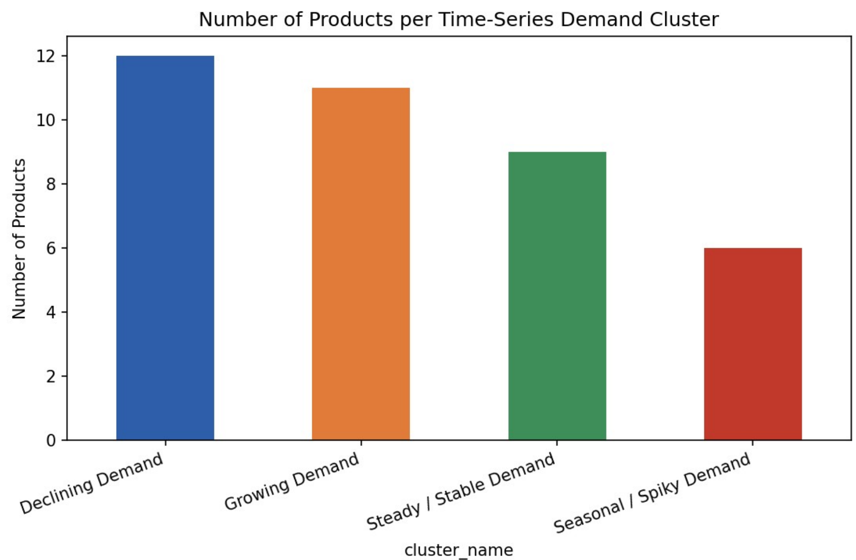

Figure 12 — Number of Products per Time-Series Demand Cluster

Observation: Declining Demand (12 products) and Growing Demand (11 products) are the two largest clusters, followed by Steady / Stable Demand (9 products) and Seasonal / Spiky Demand (6 products).

Business Interpretation: With nearly a third of the catalogue currently in the Declining Demand cluster, this is a meaningful population for the business to actively manage — either through clearance-style bundling (Chapter 7) or by pairing a declining product with a Growing Demand partner in the hope of reviving interest, rather than assuming a bundle discount alone will fix a structurally declining item.

Figure 13: Monthly Units-Sold Heatmap — Products Grouped by Time-Series Cluster

Figure 13 — Monthly Units-Sold Heatmap — Products Grouped by Time-Series Cluster

Observation:

Business Interpretation: Reading top to bottom within each cluster block, Growing Demand products visibly darken (higher units sold) toward the later months in the window, Declining Demand products lighten over the same period, and Seasonal / Spiky Demand products show isolated dark cells rather than a smooth left-to-right gradient.

Cluster	Products	Count
Steady / Stable Demand	Coffee Mug Set, Desk Organizer, Hair Serum, Laptop Cooling Pad, Phone Case, Screen Protector, Smartphone, USB-C Hub, Umbrella	9
Growing Demand	Biscuits Family Pack, Bluetooth Speaker, Gel Pen Pack, Gym Gloves, Notebook A5, Resistance Bands, Shaker Bottle, Sugar Sachets Pack, Table Lamp, Tea Leaves Premium, Wall Clock	11
Declining Demand	Conditioner 400ml, Filter Coffee Powder, Highlighter Set, Laptop 14in, Laptop Bag, Mechanical Keyboard, Milk Frother, Protein Powder 1kg, Shampoo 400ml, Sticky Notes, Water Bottle, Wireless Earbuds	12
Seasonal / Spiky Demand	Face Wash, Fast Charger 65W, Moisturizer, Sunscreen SPF50, Wireless Mouse, Yoga Mat	6

6.5 Bundle Opportunity Scoring & Recommendations

Raw association rules and cluster labels are mining outputs, not yet a merchandising decision. To convert them into an actionable, rankable list, every frequent itemset of size 2–3 was scored with a composite Bundle Opportunity Score (0–100), combining support (35% weight), maximum lift among the itemset's rules (30%), maximum confidence (20%), and estimated monthly bundle revenue (15%). Each candidate bundle was then cross-referenced against the time-series clusters of its component products.

Bundle	Score	Support	Confidence	Lift	Bundle Price	Time-Series Cluster Mix
Smartphone + Phone Case + Screen Protector	54.6	1.3%	90.6%	29.03	₹25,579	Steady + Steady + Steady
Shampoo + Conditioner	50.3	4.4%	59.7%	6.82	₹660	Declining + Declining
Laptop + Laptop Bag + Wireless Mouse	48.4	1.5%	87.0%	17.91	₹64,378	Declining + Declining + Seasonal
Filter Coffee Powder + Sugar Sachets	45.3	4.4%	48.5%	4.96	₹470	Declining + Growing
Face Wash + Moisturizer + Sunscreen SPF50	36.2	1.5%	92.9%	12.32	₹1,275	Seasonal + Seasonal + Seasonal
Laptop 14in + Laptop Bag	35.9	2.2%	44.3%	7.44	₹63,578	Declining + Declining
Protein Powder + Shaker Bottle	33.4	2.6%	62.2%	7.83	₹2,500	Declining + Growing

Figure 6: Top 10 Bundles Ranked by Composite Bundle Opportunity Score

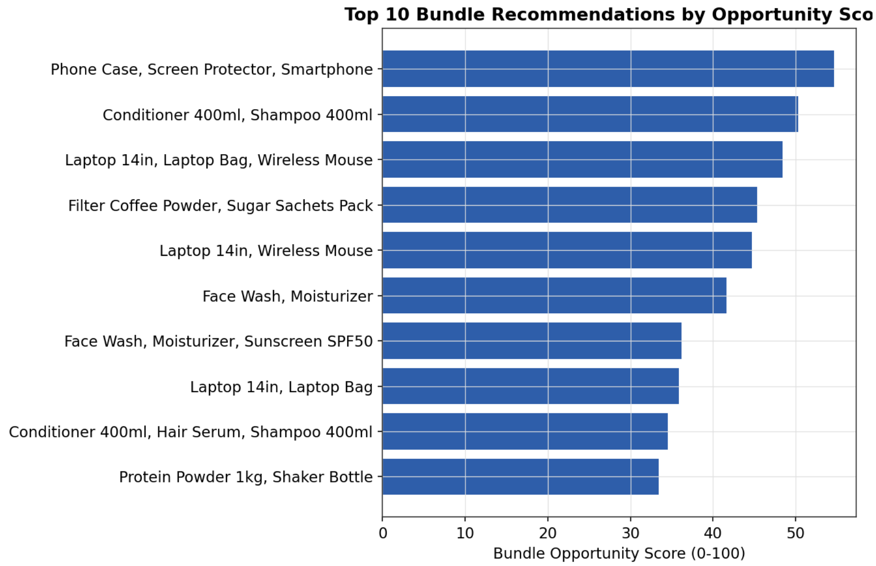

Figure 6 — Top 10 Bundles Ranked by Composite Bundle Opportunity Score

Observation: The Smartphone + Phone Case + Screen Protector bundle scores highest overall (54.6/100), followed by the Shampoo + Conditioner and Laptop accessory bundles, with scores declining smoothly rather than in sharp tiers across the top 10.

Business Interpretation: The smooth score gradient means a merchandising team does not need to treat this as a hard cutoff — all ten bundles are reasonable candidates for a first rollout, with the top 3–4 prioritized first given limited launch bandwidth.

Figure 7: Estimated Monthly Revenue Contribution per Recommended Bundle

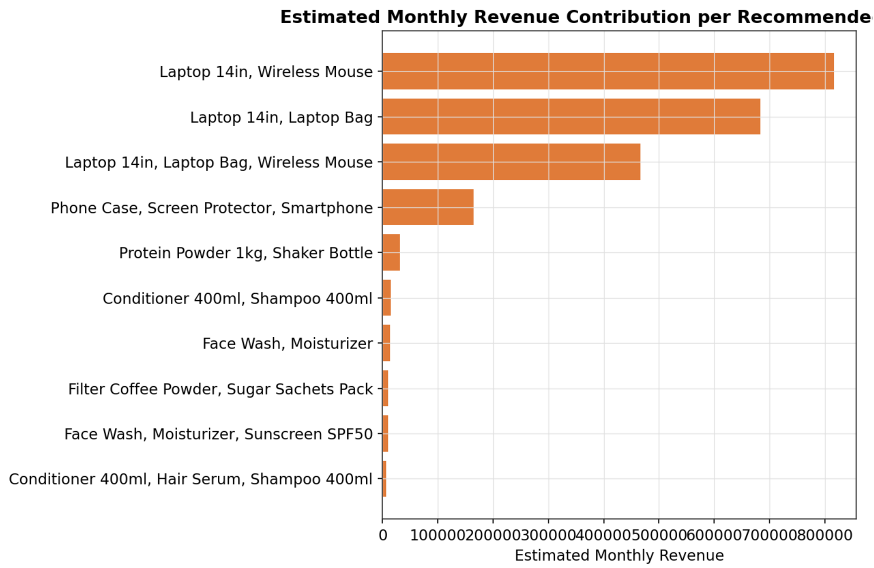

Figure 7 — Estimated Monthly Revenue Contribution per Recommended Bundle

Observation: The Laptop accessory bundles generate the highest estimated monthly revenue by a wide margin, despite not having the single highest Bundle Opportunity Score, because their component products carry a far higher unit price than the skincare, coffee, or fitness bundles.

Business Interpretation: This confirms why the scoring function in Chapter 6.5 blends statistical strength with commercial value rather than optimizing for either alone — a merchandiser optimizing purely for revenue would prioritize differently than one optimizing purely for statistical confidence, and the composite score is designed to balance both perspectives.

6.6 Summary

This chapter combined two independent analytical lenses — market basket analysis and time-series clustering — into a single, ranked, and priced bundle recommendation list. The highest-scoring bundle (Smartphone + Phone Case + Screen Protector) is strong on every dimension: high lift, high confidence, and all three components in the same Steady / Stable Demand cluster. Other bundles, such as Shampoo + Conditioner and Laptop 14in + Laptop Bag, are statistically strong (high confidence and lift) but pair two Declining Demand products, which Chapter 7 discusses as a signal for a different merchandising strategy (clearance-style bundling) rather than a straightforward full-price bundle launch.

CHAPTER 7: BUSINESS INSIGHTS AND RECOMMENDATIONS

7.1 Introduction

This chapter translates the statistical findings of Chapters 5 and 6 into concrete, actionable business insights and recommendations for a merchandising or e-commerce operations team.

7.2 Key Business Insights

- Nearly two-thirds of orders (65.6%) already contain more than one item, indicating that customers are naturally inclined toward multi-item purchases — bundle merchandising can convert this existing behavior into a structured, monetized offer rather than creating new behavior from scratch.
- The single strongest pattern in the dataset (Phone Case + Screen Protector → Smartphone, 29x lift) is an accessory-attachment pattern — customers buying phone accessories have effectively already decided to buy the phone — and all three components sit in the Steady / Stable Demand time-series cluster, making this the safest, highest-confidence bundle to launch first.
- High-ticket clusters (laptop + accessories) generate the largest estimated absolute revenue contribution per bundle despite moderate order counts, because of their high unit price — but two of the three laptop-bundle components are in the Declining Demand cluster, so this bundle should be monitored closely and may need a deeper discount to sustain revenue rather than assuming continued organic demand.
- Several statistically strong bundles (Shampoo + Conditioner, Laptop + Laptop Bag) pair two Declining Demand products together — bundling alone is unlikely to reverse a genuine downward demand trend, and these candidates are better suited to a clearance/inventory-reduction framing than a premium bundle framing.
- Bundles that pair a Declining Demand product with a Growing Demand product (Filter Coffee Powder + Sugar Sachets Pack; Protein Powder + Shaker Bottle) represent an interesting “revival” opportunity — the growing partner's rising visibility may help lift the declining product's exposure, an effect that would be worth measuring directly once such a bundle is live.
- 88.7% of customers are repeat purchasers, with an average of 3.44 orders per customer over the year — a large enough repeat base to support customer-specific, purchase-history-driven bundle recommendations in addition to store-wide bundles.

7.3 Business Recommendations

- Launch the Smartphone + Phone Case + Screen Protector bundle first: it has the highest composite score, the strongest lift and confidence, and all components are in the Steady / Stable Demand cluster, minimizing execution risk.
- Position the laptop accessory bundle (Laptop 14in + Laptop Bag + Wireless Mouse) as a premium, moderately-discounted (13–15%) bundle given its high revenue contribution, but track its attach rate closely given its Declining Demand components.
- Reframe the Shampoo + Conditioner and Laptop + Laptop Bag two-item bundles as clearance or loyalty-reward bundles rather than premium launches, given that both pair two Declining Demand products.

- Pilot a “revival bundle” combining a Declining Demand product with its Growing Demand market-basket partner (Filter Coffee Powder + Sugar Sachets Pack is the strongest such candidate) and measure whether the declining product's own standalone sales improve over the following quarter.
- Re-run both the Apriori mining and the time-series clustering pipeline on a monthly or quarterly cadence, since co-purchase patterns and demand trajectories will both shift over time — a bundle that scores well this quarter may not next quarter, and a Declining Demand product this quarter could stabilize or recover.

7.4 Business Value of the Project

This project demonstrates that bundle recommendation does not need to remain a matter of merchandiser intuition — it is a well-posed, tractable data mining problem, and combining market basket analysis with time-series clustering produces a materially richer, more actionable recommendation than either technique alone would. For a real e-commerce deployment, these recommendations could be operationalized through a platform's existing bundle-management feature. As a concrete reference point, the open-source Sylius e-commerce framework's SyliusProductBundlePlugin already provides the data model needed to receive such recommendations: a bundle is represented as an ordinary Product carrying an attached ProductBundle configuration of component ProductVariants and quantities. A scheduled export of Sylius order data could feed this project's analytics pipeline, and approved bundle recommendations could be written back into the store through the same ProductBundle and ProductBundleItem entities the plugin already defines, with a human merchandiser remaining in the approval loop before anything goes live on the storefront.

CHAPTER 8: CONCLUSION, LIMITATIONS, FUTURE SCOPE AND REFERENCES

8.1 Conclusion

This project set out to answer a question that most e-commerce platforms leave to merchandiser intuition: not how to sell a bundle, but which bundles are worth building in the first place, and when. By combining a from-scratch Apriori implementation for market basket analysis with a from-scratch k-means implementation for time-series clustering, this project built a complete, auditable analytics pipeline that converts raw transactional data into a ranked, priced, and demand-aware list of bundle recommendations. The pipeline was validated against a synthetic but carefully engineered dataset, correctly recovering all ten deliberately designed co-purchase clusters as its top recommendations, with the strongest pattern (phone + accessories) reaching a lift of 29x and a confidence of over 90%, and correctly separating that bundle's components into the Steady / Stable Demand time-series cluster distinct from lower-priority Declining Demand bundle candidates.

8.2 Limitations

- The dataset used in this project is synthetic. While it was carefully engineered to reproduce realistic co-purchase and seasonal structure, results on a real store's transaction data may differ in magnitude, and threshold parameters (minimum support, minimum confidence, number of time-series clusters) would need to be re-tuned against real data volumes.
- The time-series clustering model uses only 12 monthly data points per product, which limits the ability to distinguish genuine seasonal cycles from short-term noise; a multi-year dataset would allow more confident seasonal-pattern detection.
- The Bundle Opportunity Score's weighting (35% support, 30% lift, 20% confidence, 15% revenue) was set by the analyst's judgement rather than optimized against a labeled set of historically successful bundles, since no such historical bundle-performance data was available for this project.
- The analysis does not yet incorporate live A/B testing of the recommended bundles on an actual storefront, so the revenue-impact estimates in Chapter 6 remain projections rather than measured outcomes.

8.3 Future Scope

- Validate the pipeline against real, anonymized e-commerce order exports rather than synthetic data, and re-tune all thresholds to the store's actual order volume and catalogue size.
- Extend the time-series clustering component with multi-year data to reliably distinguish true seasonal cycles from single-year noise.
- Incorporate sequential/time-aware market basket patterns (e.g., a customer who bought a printer last month is now likely to buy ink) using sequence mining techniques, rather than only same-basket co-occurrence.
- Add automatic A/B measurement of live bundle performance (actual attach rate, actual lift in average order value) feeding directly back into the Bundle Opportunity Score's weighting.
- Build the merchandiser review and approval workflow as an actual lightweight internal tool or a custom admin panel screen (e.g. within Sylius), rather than a manual process, to reduce the operational overhead of adopting this system.

- Explore alternative clustering distance measures such as Dynamic Time Warping (DTW) in place of Euclidean distance, which may better capture demand patterns that are similarly shaped but shifted in time.

8.4 References

- Agrawal, R. and Srikant, R. (1994). “Fast Algorithms for Mining Association Rules.” Proceedings of the 20th International Conference on Very Large Data Bases (VLDB).
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